

NHS Data Analysis – Technical Report

Background

The NHS must understand the utilisation trends of components in its network so they can budget correctly when considering infrastructure and resource changes to match an increasing population. To better understand the issues, they have provided historical data and asked 2 questions, which I have broken down:

1. Has there been adequate staff and capacity in the networks?
 - a. Is the mix of staff correct?
 - b. Is capacity utilised consistently?
 - c. How do DNA's affect capacity?
 - d. Will increasing patient numbers affect capacity?
2. What was the actual utilisation of resources?
 - a. Are some resources overstretched while others are barely used?
 - b. Can we make better use of all resources to ease pressure points?
 - c. Can we do anything to reduce the number of DNA's?

Resources

<https://www.nhs.uk/nhs-services/find-your-local-integrated-care-board/>

<https://www.lancashireandsouthcumbria.icb.nhs.uk/about-us/our-vision-objectives-and-priorities>

<https://www.transformationpartners.nhs.uk/wp-content/uploads/2017/12/Guide-for-promoting-extended-access-services-at-practices-November-2018.pdf>

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Analytical Approach

The data provided by the NHS:

<u>Data Provided by NHS</u>			
National Categories			
Information	Breakdown By	Added to CSV	Added in Python
Service Setting	Appointment Date	Region Code	Region Name
Context Type	Appointment Month		ICB Name
National Category	ICB Code		
Appointment Count	Sub-ICB Location		
Appointments Regional			
Information	Breakdown By	Added to CSV	Added in Python
Appointment Status	Appointment Month	Region Code	Region Name
HCP Type	ICB Code		ICB Name
Appointment Mode			Appointment Date
Time Between Book & Appt			
Appointment Count			
Actual Duration			
Information	Breakdown By	Added to CSV	Added in Python
Actual Duration	Appointment Date		Region Name
Appointment Count	Region Code		
	ICB Code		
	Sub-ICB Location		
Tweets			
Information	Unused (removed in Python)	Reason	
tweet_full_text	tweet_id	Unique ID for the tweet, not relevant	
tweet_entities_hashtags	tweet_entities	Only useful info already in other fields	
tweet_retweet_count	tweet_metadata	Same for every record	
tweet_favorite_count	tweet_favorited	Same for every record	
	tweet_retweeted	Same for every record	
	tweet_lang	Same for every record	

Region codes and names were added to all csv's to allow grouping by region and to add detail. ICB names were added to National Categories and Appointments Regional for descriptive purposes.

Appointment Date (all set to 1st of month) was added to Appointments Regional in Python for continuity. Date strings were converted to datetime as necessary.

It was also noted previously that there is no date in appointments_regional, there is only the month.
For consistency, I have added a date to this dataframe as well, although I have had to make the day of the date be the 1st of the month.

```
# There isn't an appointment_date field in ar!
# Adding one here, but all the dates are going to show the first of the appointment month
ar['appointment_date'] = pd.to_datetime(ar['appointment_month'])

# View the DataFrame.
ar
```

	region_ons_code	icb_ons_code	appointment_month	appointment_status	hcp_type	appointment_mode	time_between_book_and_appointment	count_of_appointments	appointment_date
0	E40000005	E54000034	2020-01	Attended	GP	Face-to-Face	1 Day	8107	2020-01-01
1	E40000005	E54000034	2020-01	Attended	GP	Face-to-Face	15 to 21 Days	6791	2020-01-01

Appointment date in actual duration is held as a string rather than a date, whereas the date in national categories is a datetime (as noted previously).

```
# Change the date format of ad['appointment_date'].
ad['appointment_date'] = pd.to_datetime(ad['appointment_date'], format='%d-%b-%y')

# View the DataFrame.
ad
```

	sub_icb_location_code	sub_icb_location_ons_code	sub_icb_location_name	icb_ons_code	region_ons_code	appointment_date	actual_duration	count_of_appointments
0	00L	E38000130	NHS North East and North Cumbria ICB - 00L	E54000050	E40000012	2021-12-01	31-60 Minutes	364
1	00L	E38000130	NHS North East and North Cumbria ICB - 00L	E54000050	E40000012	2021-12-01	21-30 Minutes	619

No other data cleaning was required. There are no NULL entries in the data. The data in each file has aggregated appointment counts, different date ranges, different levels of granularity and no unique identifiers so cannot be merged.

To determine if the appointment data matches in all datasets, I created a new csv file and added calculations to get the DNA and Unknown breakdowns for comparison.

I want to determine if the number of appointments across data sets matches up, and to see if we can tell if the unknown statuses in National Categories are attended or DNA.

For this, I created a pivoted data set in Excel from National Categories to get the appointments by ICB for the common date range (01/12/2021 - 30/06/2022), then did the same for Actual Duration and Appointments Regional for appointments in the same date range.

I added these two sets of results to a new csv file (appts_Dec21_Jun22.csv) - This was all done in Excel.

```
[39]: appts_data = pd.read_csv('appts_Dec21_Jun22.csv')
appts_data
```

	icb_ons_code	ar_DNA	ar_Unknown	ar_Attended	ad_appts	ar_D+U_Total	nc_ad_diff	nc_appts	ar_total
0	E54000008	430735	353458	7344348	7344348	784193	784193	8128541	8128541
1	E54000010	152077	139442	2954384	2954384	291519	291519	3245903	3245903
2	E54000011	60862	59170	1459819	1459819	120032	120032	1579851	1579851
3	E54000013	96582	85887	2457468	2457468	182469	182469	2639937	2639937
4	E54000015	146749	116851	3610315	3610315	263600	263600	3873915	3873915
5	E54000018	130203	112689	2751699	2751699	242892	242892	2994591	2994591
6	E54000019	116872	89345	2696703	2696703	206217	206217	2902920	2902920
7	E54000022	137089	124966	3490647	3490647	262055	262055	3752702	3752702
8	E54000023	113092	102341	3059758	3059758	215433	215433	3275191	3275191

From the resulting dataset, I can see that:

1. The attended appointments in appointments_regional (ar_Attended) match the count of appointments in actual_duration (ad_appts)
2. The DNA + Unknown appointments in appointments_regional (ar_D+U_Total) match the difference in number of appointments between national_categories and actual_duration (nc_ad_diff)
3. The count of appointments in national_categories (nc_appts) matches the total count of appointments in appointments_regional (ar_total)

This suggests that:

- All appointments marked as attended (in appointments_regional) have a duration (in actual_duration)
- National Categories and Appointments Regional are the same appointments for that date range with different aggregations.
- The appointments in actual_duration are the attended appointments in the other two datasets.

This would also suggest that appointments marked as unknown in appointments_regional are actually DNA's. However, due to concerns about data quality, I am reluctant to say this for certain as it may be that they are unknown simply because the data in actual_duration is incomplete. I am fairly confident that all data sets contain information about the same appointments and it's not a random sample in each set.

I used this new data to create an additional CSV file including patient numbers using additional data from here: <https://digital.nhs.uk/data-and-information/publications/statistical/patients-registered-at-a-gp-practice/>.

For the next section, I took patient data from the [NHS Website](#), from which I downloaded 9 csv files for patient data from 1st December 2021 to 1st July 2022 (example 01-Jun-22: [File Download](#)), and created additional csv's in Excel to explore patient data versus appointments.

For this, I created a new csv file (icb_location_patients_appointments_Dec21_to_Jun22.csv), which takes the pivoted data created in the new apts_dec21_Jun22.csv appointments data, and added the downloaded data from the gp_reg_pat_prac_all.csv files for each month.

I have concentrated on the patient data for the beginning and end of the period in the following investigation.

```
[64]: # Import and sense-check the data
ilpa = pd.read_csv('icb_location_patients_appointments_Dec21_to_Jun22.csv')

# View the DataFrame.
ilpa
```

	region	region_ons_code	icb	icb_ons_code	Appt Count	Attended	DNA	Unknown	D+U	PatientCount_Dec21	PatientCount_Jan22	PatientCount_Feb22	PatientCount_Mar22
0	London	E40000003	North West London	E54000027	7631623	6976986	373794	280843	654637	2741383	2751843	2757592	2765225
1	London	E40000003	North Central London	E54000028	4183041	3795250	234642	153149	387791	1737523	1742487	1745349	1747362
2	London	E40000003	North East London	E54000029	5967026	5341883	362285	262858	625143	2339566	2346009	2350561	2355246
3	London	E40000003	South East London	E54000030	4839805	4360079	272101	207625	479726	2044900	2047004	2049910	2050935
4	London	E40000003	South West London	E54000031	4443408	4014321	249503	179584	429087	1724104	1726248	1727718	1728778
	South												

Melted data set for graph plotting:

```
# Get the earliest and latest patient counts from the data, and include the appointment and attended figures.
ilpa_regions = ilpa.groupby('region') \
[["PatientCount_Dec21", "PatientCount_Jul22", "Appt Count", "Attended"]] \
.sum() \
.reset_index()

# Melt the data for plotting
ilpa_regions_melt = ilpa_regions.melt(id_vars='region', \
value_vars=["PatientCount_Dec21", "PatientCount_Jul22", "Appt Count", "Attended"], \
var_name='Attribute', \
value_name='Value').reset_index()

ilpa_regions_melt
```

	index	region	Attribute	Value
0	0	East of England	PatientCount_Dec21	7073995
1	1	London	PatientCount_Dec21	10587476
2	2	Midlands	PatientCount_Dec21	11489282
3	3	North East and Yorkshire	PatientCount_Dec21	9012291
4	4	North West	PatientCount_Dec21	7696530
5	5	South East	PatientCount_Dec21	9553164
6	6	South West	PatientCount_Dec21	5983101
7	7	East of England	PatientCount_Jul22	7104254
8	8	London	PatientCount_Jul22	10697223
9	9	Midlands	PatientCount_Jul22	11599656
10	10	North East and Yorkshire	PatientCount_Jul22	9035991
11	11	North West	PatientCount_Jul22	7710356
12	12	South East	PatientCount_Jul22	9609875
13	13	South West	PatientCount_Jul22	6011587
14	14	East of England	Appt Count	21008711
15	15	London	Appt Count	27064903
16	16	Midlands	Appt Count	35378427
17	17	North East and Yorkshire	Appt Count	29675995

I created 2 CSV files to explore DNA percentages by appointment mode.

For this I loaded the data into SQL and created queries to extract the data to CSV. I split appointment_month into Month and Year in SQL for easier use

SQL Queries and CSV's are provided separately.

```
WITH ar AS
(
  SELECT icb_ons_code, appointment_mode, appointment_status, count_of_appointments, appointment_month, appointment_year
  FROM appointments_regional
  WHERE appointment_year = CASE WHEN appointment_month > 7 THEN 2021 ELSE 2022 END
),
appt_data AS
(
  SELECT icb_ons_code, appointment_mode, SUM(count_of_appointments) AS total_appointments, appointment_month, appointment_year
  FROM appointments_regional
  GROUP BY icb_ons_code, appointment_mode, appointment_month, appointment_year
)

SELECT ar.icb_ons_code,
ar.appointment_mode,
ar.appointment_status,
SUM(CAST(ar.count_of_appointments AS DECIMAL(10,2))) / CAST(total_appointments AS DECIMAL(10,2)) * 100.00 AS percent_dna_appts,
CONCAT_WS('-', ar.appointment_year, ar.appointment_month) AS appointment_month
FROM ar
INNER JOIN appt_data
ON ar.icb_ons_code = appt_data.icb_ons_code AND
ar.appointment_mode = appt_data.appointment_mode AND
ar.appointment_month = appt_data.appointment_month AND
ar.appointment_year = appt_data.appointment_year
GROUP BY ar.icb_ons_code, ar.appointment_mode, ar.appointment_status, total_appointments, ar.appointment_month, ar.appointment_year
HAVING ar.appointment_status IN ('DNA', 'Unknown')
ORDER BY percent_dna_appts DESC
```

Here we are looking at the percentage of DNA and unknown per appointment type, and comparing the top 5 ICB's with the bottom 5 ICB's by count of appointments.

```
# Load the csv file and merge in the icb names
ar_modes = pd.read_csv('dna_unknown_Aug21_to_Jun22_monthly.csv')
ar_modes_icb = pd.merge(ar_modes, icbs, on='icb_ons_code')

ar_modes_icb.head()
```

	icb_ons_code	appointment_mode	appointment_status	percent_dna_appts	appointment_month		icb_index
0	E54000028	Unknown	Unknown	100.000000	2022-06	North Central London	1
1	E54000028	Unknown	Unknown	100.000000	2022-05	North Central London	1
2	E54000018	Home Visit	Unknown	49.691665	2021-12	Coventry and Warwickshire	1
3	E54000018	Home Visit	Unknown	48.808457	2022-03	Coventry and Warwickshire	1
4	E54000044	Video/Online	Unknown	48.210800	2021-09	Buckinghamshire Oxfordshire and Berkshire West	1

Filter the data to just look at DNA appointments

```
ar_dna = ar_modes_icb[ar_modes_icb['appointment_status'] == 'DNA'].sort_values(by=['icb', 'appointment_month'], ascending=[False, True])
ar_dna.head()
```

	icb_ons_code	appointment_mode	appointment_status	percent_dna_appts	appointment_month		icb_index
1755	E54000054	Face-to-Face	DNA	4.550261	2021-08	West Yorkshire	1
2293	E54000054	Video/Online	DNA	3.636364	2021-08	West Yorkshire	1
3145	E54000054	Unknown	DNA	2.605779	2021-08	West Yorkshire	1
3763	E54000054	Home Visit	DNA	1.919339	2021-08	West Yorkshire	1
4327	E54000054	Telephone	DNA	0.873229	2021-08	West Yorkshire	1

As a follow-up, I am now looking at DNA's for face-to-face and telephone appointments, for the top and bottom ICB's as a whole.

```
# Get the data from the csv file created earlier and merge in the ICB names
du = pd.read_csv('dna_unknown_Aug21_to_Jun22.csv')
du_icb = pd.merge(du, icbs, on='icb_ons_code')

# View the DataFrame
du_icb.head()
```

	icb_ons_code	appointment_mode	appointment_status	percent_dna_appts	icb	index
0	E54000018	Home Visit	Unknown	20.004231	Coventry and Warwickshire	1
1	E54000010	Home Visit	Unknown	14.129862	Staffordshire and Stoke-on-Trent	1
2	E54000062	Home Visit	DNA	13.510837	Black Country	1
3	E54000031	Home Visit	Unknown	12.701258	South West London	1
4	E54000030	Home Visit	Unknown	10.283646	South East London	1

```
# Create a subset of the data containing only the top and bottom ICB's and Face-to-face and telephone appointments
du_icb_subset = du_icb[((du_icb['icb'].isin(top_icbs)) | (du_icb['icb'].isin(low_icbs))) & \
    (du_icb['appointment_status'] == 'DNA') & \
    (du_icb['appointment_mode'].isin(['Face-to-Face', 'Telephone']))]

# View the subset
du_icb_subset
```

	icb_ons_code	appointment_mode	appointment_status	percent_dna_appts	icb	index
48	E54000057	Face-to-Face	DNA	3.402324	Greater Manchester	1
63	E54000034	Face-to-Face	DNA	3.045798	Frimley	1
69	E54000008	Face-to-Face	DNA	2.876439	Cheshire and Merseyside	1
78	E54000027	Face-to-Face	DNA	2.740693	North West London	1
98	E54000038	Face-to-Face	DNA	2.469407	Somerset	1
117	E54000050	Face-to-Face	DNA	2.216971	North East and North Cumbria	1
137	E54000043	Face-to-Face	DNA	2.011029	Gloucestershire	1
138	E54000054	Face-to-Face	DNA	2.007540	West Yorkshire	1
144	E54000011	Face-to-Face	DNA	1.969831	Shropshire Telford and Wrekin	1
145	E54000036	Face-to-Face	DNA	1.967775	Cornwall and The Isles Of Scilly	1
236	E54000008	Telephone	DNA	1.296656	Cheshire and Merseyside	1
241	E54000038	Telephone	DNA	1.264469	Somerset	1
242	E54000057	Telephone	DNA	1.250474	Greater Manchester	1
313	E54000034	Telephone	DNA	0.930199	Frimley	1
333	E54000027	Telephone	DNA	0.875466	North West London	1
360	E54000011	Telephone	DNA	0.730222	Shropshire Telford and Wrekin	1
365	E54000050	Telephone	DNA	0.680170	North East and North Cumbria	1

Any additional merging, filtering and grouping of data was done in python as required for the task at hand and is documented in the notebook.

Data Limitations

1. Too many unknowns and data quality issues in the data provided.
2. Differences in granularity between datasets made comparisons at lower levels difficult.
3. Some data may not be typical due to changes in service delivery during COVID, potentially higher patient numbers, lockdown restrictions and other government measures.
4. Aggregation of “Other Practice Staff” makes it difficult to see what HCP’s are taking on appointments.
5. Context types are not useful in their current form.
6. Patient demographics may have been useful for determining DNA reasons, this was not provided (understandably, for GDPR reasons).
7. The Twitter data collected had a lot of redundant fields, did not appear to have anything of relevance, was mostly about USA healthcare, and did not have a date range. I did not use this for analysis.

Data Recommendations:

The various systems used by GP’s must be consistent and record accurately. Encourage GP’s and other practice staff to use data collection systems properly, even during busy periods. Ease of use is paramount.

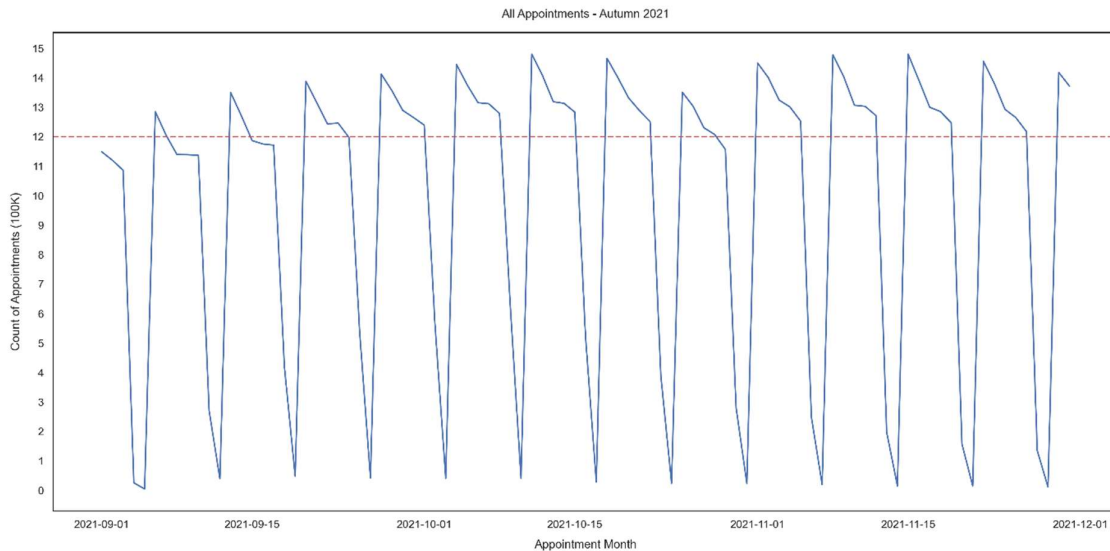
Remove the aggregation of “Other Practice Staff” to better see the HCP details to help with planning.

Collection of social media needs to be better thought out:

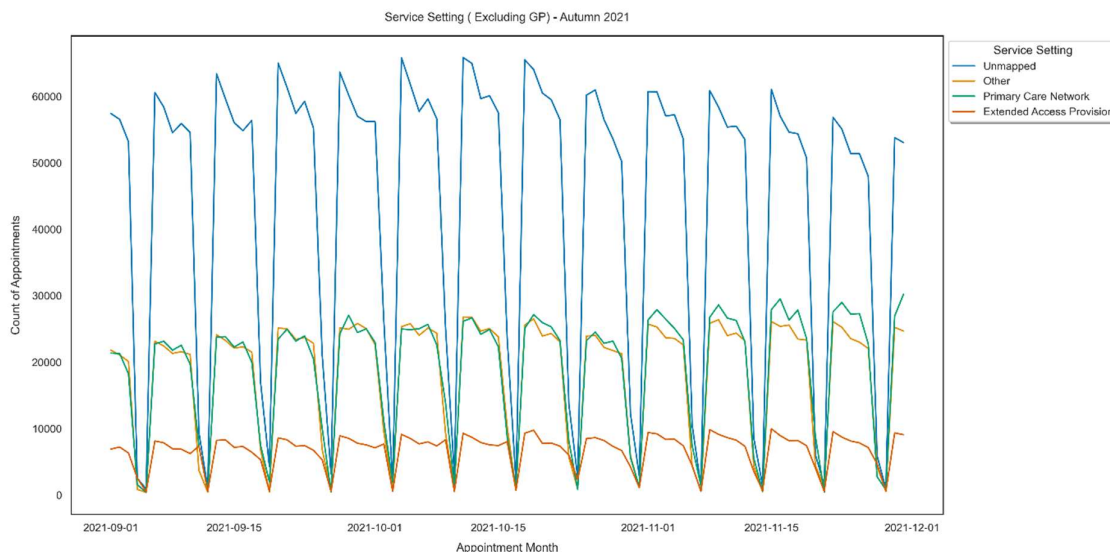
- Use appropriate hashtags such as #NHS
- Create polls and surveys
- Embed Twitter reels into NHS websites (encourage discussion / participation)

Visualisations and Insights

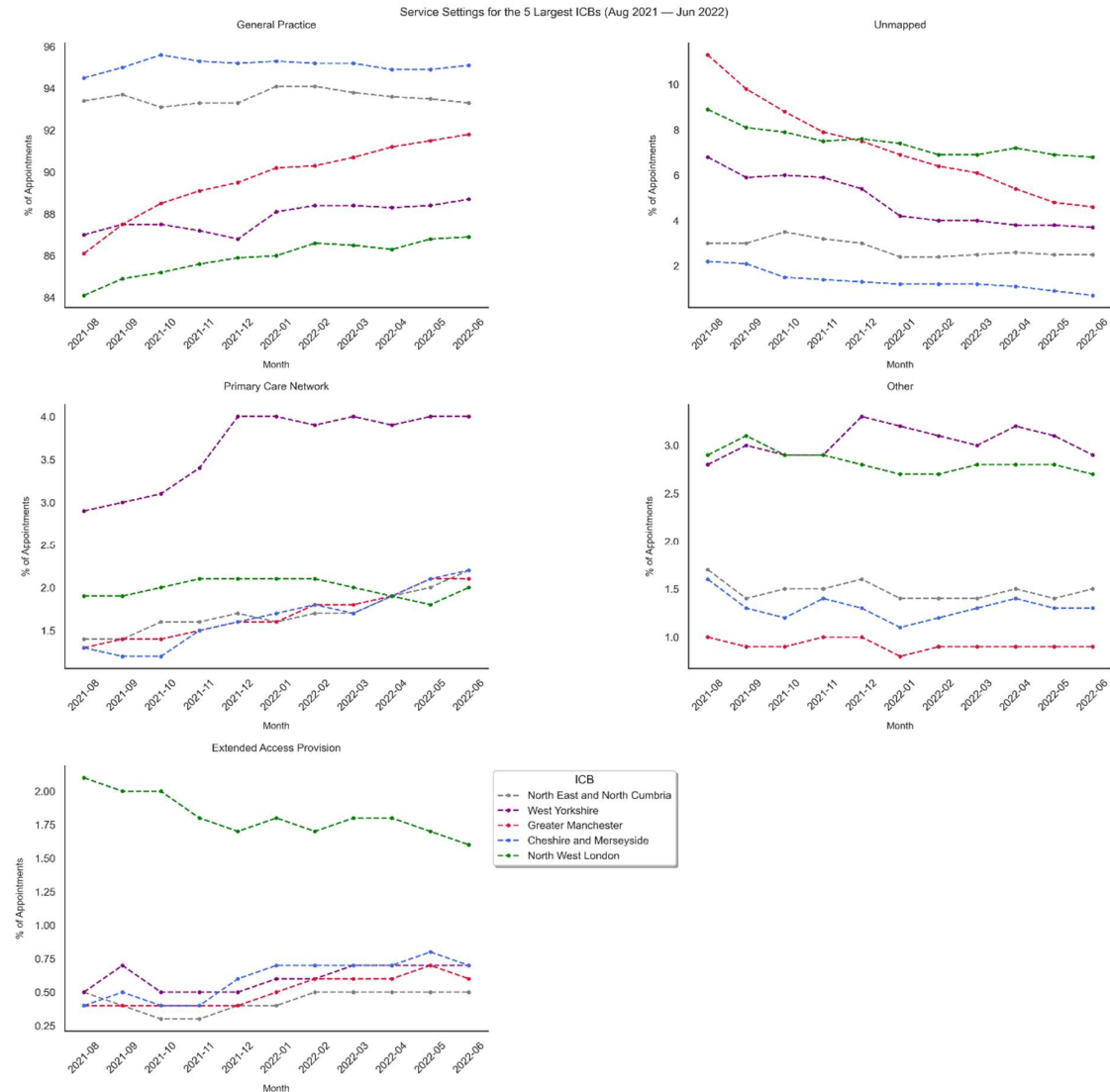
Exploratory visualisations to gather insights into service setting utilisation. Red line indicates an assumed theoretical maximum capacity of 1.2 million appointments per day. Autumn 2021 used (this was the busiest period). Maximum capacity exceeded constantly, Mondays and Tuesdays are the worst days. There is a sudden drop at weekends indicating little provision of service.



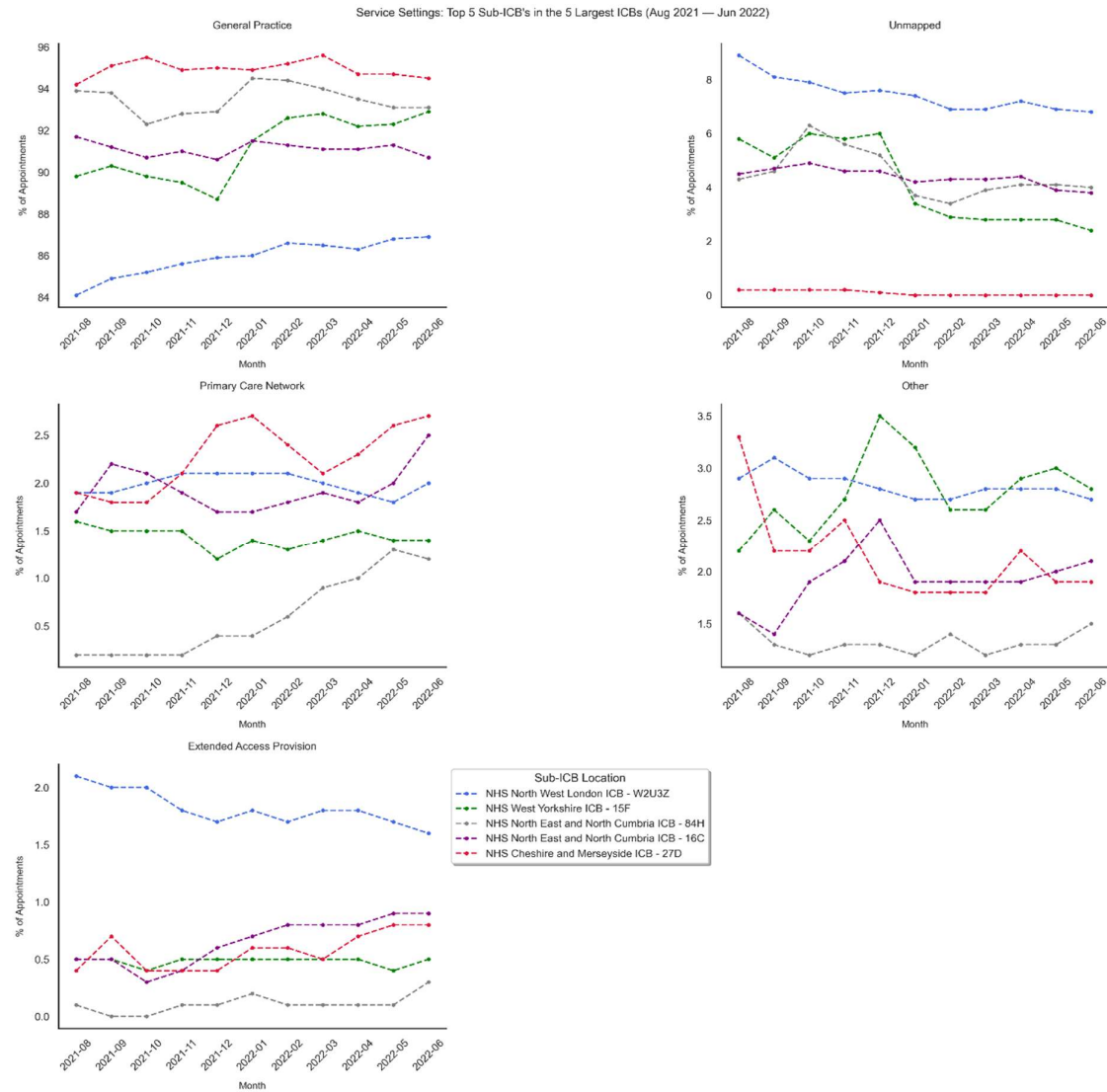
Service setting utilisation other than General Practice is low. Extended Access more so than the others. Extended Access utilisation at weekends needs to be much higher. This may ease demand at the beginning of the work week. Large number of unknowns.



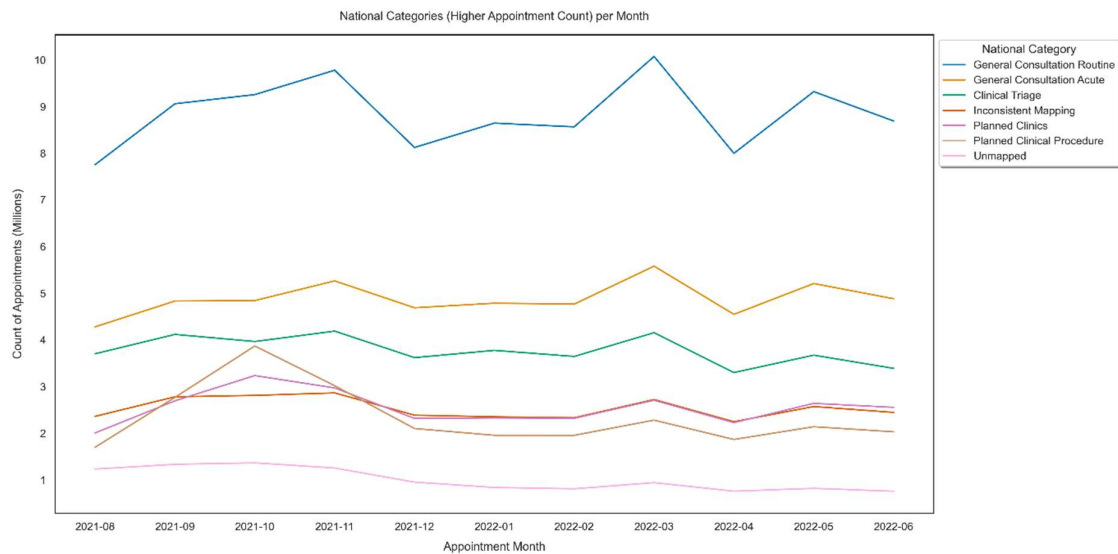
Service Setting in more detail for the top 5 ICB's over time to see patterns of usage. North-West London and West Yorkshire making better use of Extended Access and Other services has reduced pressure on GP services. Reducing usage of Extended Access adds more pressure to General Practice.



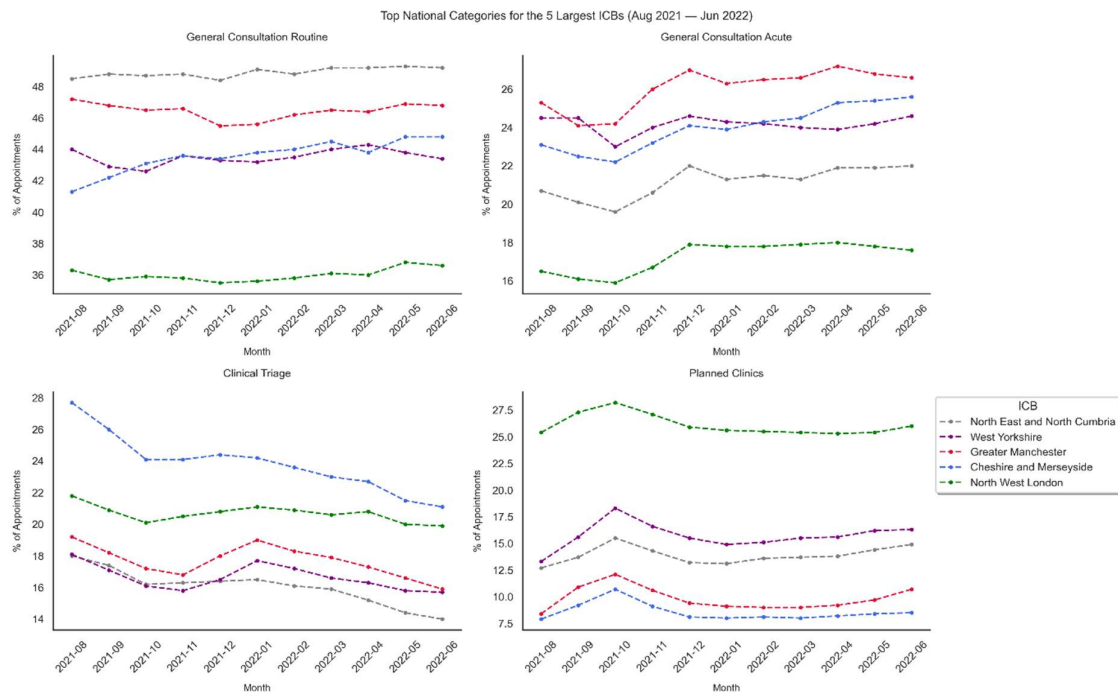
Similar patterns can be seen at sub-location level too. Cheshire and Merseyside using Primary Care Networks has had a similar effect to North London using Extended Access, indicating usage of either or both will help with General Practice bottlenecks.



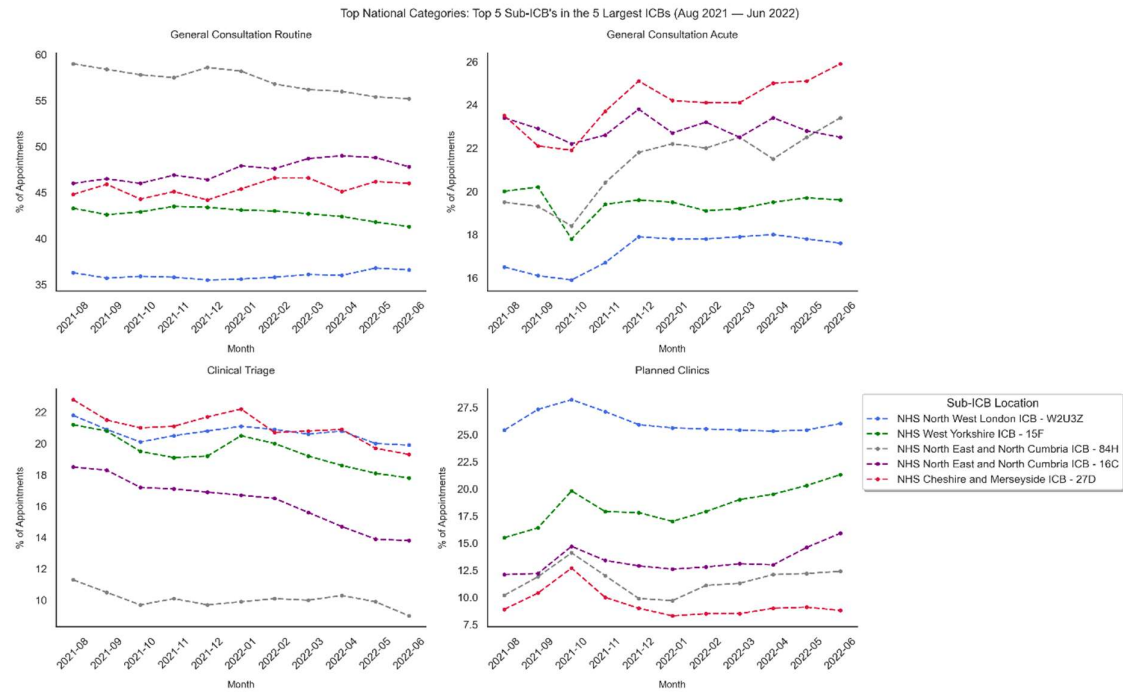
National Category exploratory visualisation to establish the top categories in use.



At ICB level, we can see that North-West London's use of planned clinics and clinical triage has reduced General Consultation appointments. Cheshire and Merseyside also use clinical triage, but this has not had the same effect, suggesting that while NW London triage patients to planned clinics, Cheshire and Merseyside are still sending patients on a General Consultation pathway.



A similar pattern can be seen at sub-location level. NHS West Yorkshire appear to be better at clinical triage than other sub-locations in the same ICB and have also lowered their general practice appointments in favour of planned clinics.



To investigate capacity, I created a graph to show the difference in capacity over time considering just weekdays, weekdays including Saturdays, and all days.

I first had to determine the number of each day type in each month, and excluded bank holidays, Easter and Christmas from the results

Start by looking at the composition of weekdays and weekend days in each month

```
# Create a dataframe for the months we want to view
month_weekdays = pd.DataFrame({'appointment_month': ['2021-08', '2021-09', '2021-10', '2021-11', '2021-12', '2022-01', '2022-02', '2022-03', '2022-04', '2022-05', '2022-06']})

# Get the months as a list of strings and a list of datetimes
month_string_list = month_weekdays['appointment_month']
month_list = pd.to_datetime(month_weekdays['appointment_month'])

# Create 2 lists for month 1 and month 2
month_1 = month_list.to_numpy().astype('datetime64[M]')
month_2 = month_1 + np.array([1], dtype='timedelta64[M]')

# Calculate the number of weekdays, Saturdays, Sundays and total days between month 1 and month 2

# Get holiday dates:
# Christmas (25-28 Dec) - Christmas day was a Saturday so Mon & Tue were bank holidays,
# New Year (1-3 Jan) - New Year's day was a Saturday so included 2nd (sun) and 3rd (Mon) as holidays
# Easter (15-18 Apr)
# Bank hols (2nd May, 2nd & 3rd Jun) - there was an extra Jubilee day bank holiday in June 2022
holiday_wd = ['2021-08-30', '2021-12-27', '2021-12-28', '2022-01-03', '2022-04-15', '2022-04-18', '2022-05-02', '2022-06-02', '2022-06-03']
holiday_sat = ['2021-12-25', '2022-01-01', '2022-04-16']
holiday_sun = ['2021-12-26', '2022-01-02', '2022-04-17']

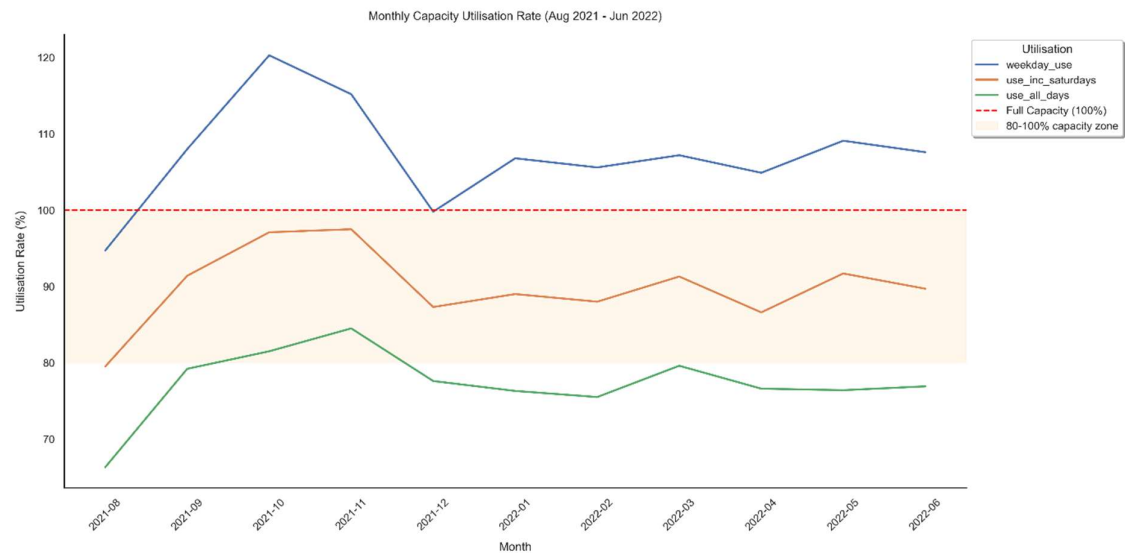
# Add the day calculations as new columns in the dataframe (exclude bank holidays etc. for all)
month_weekdays['weekdays'] = np.busday_count(month_1, month_2, holidays=holiday_wd)
month_weekdays['saturdays'] = np.busday_count(month_1, month_2, weekmask='Sat', holidays=holiday_sat)
month_weekdays['sundays'] = np.busday_count(month_1, month_2, weekmask='Sun', holidays=holiday_sun)
month_weekdays['total_days'] = month_weekdays['weekdays'] + month_weekdays['saturdays'] + month_weekdays['sundays']

# Convert appointment_month to string
month_weekdays['appointment_month'] = month_weekdays['appointment_month'].astype(str)

# View the dataframe
month_weekdays
```

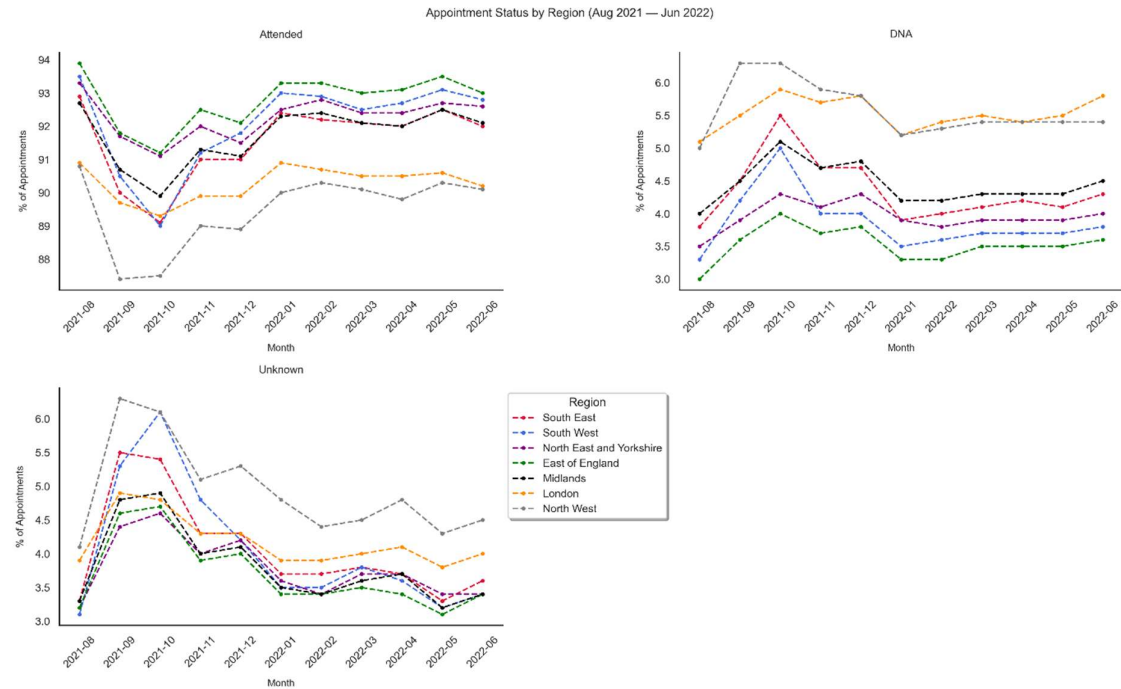
	appointment_month	weekdays	saturdays	sundays	total_days
0	2021-08	21	4	5	30
1	2021-09	22	4	4	30
2	2021-10	21	5	5	31
3	2021-11	22	4	4	30
4	2021-12	21	3	3	27
5	2022-01	20	4	4	28
6	2022-02	20	4	4	28
7	2022-03	23	4	4	31
8	2022-04	19	4	3	26
9	2022-05	21	4	5	30
10	2022-06	20	4	4	28

This has then been plotted onto a graph as shown:



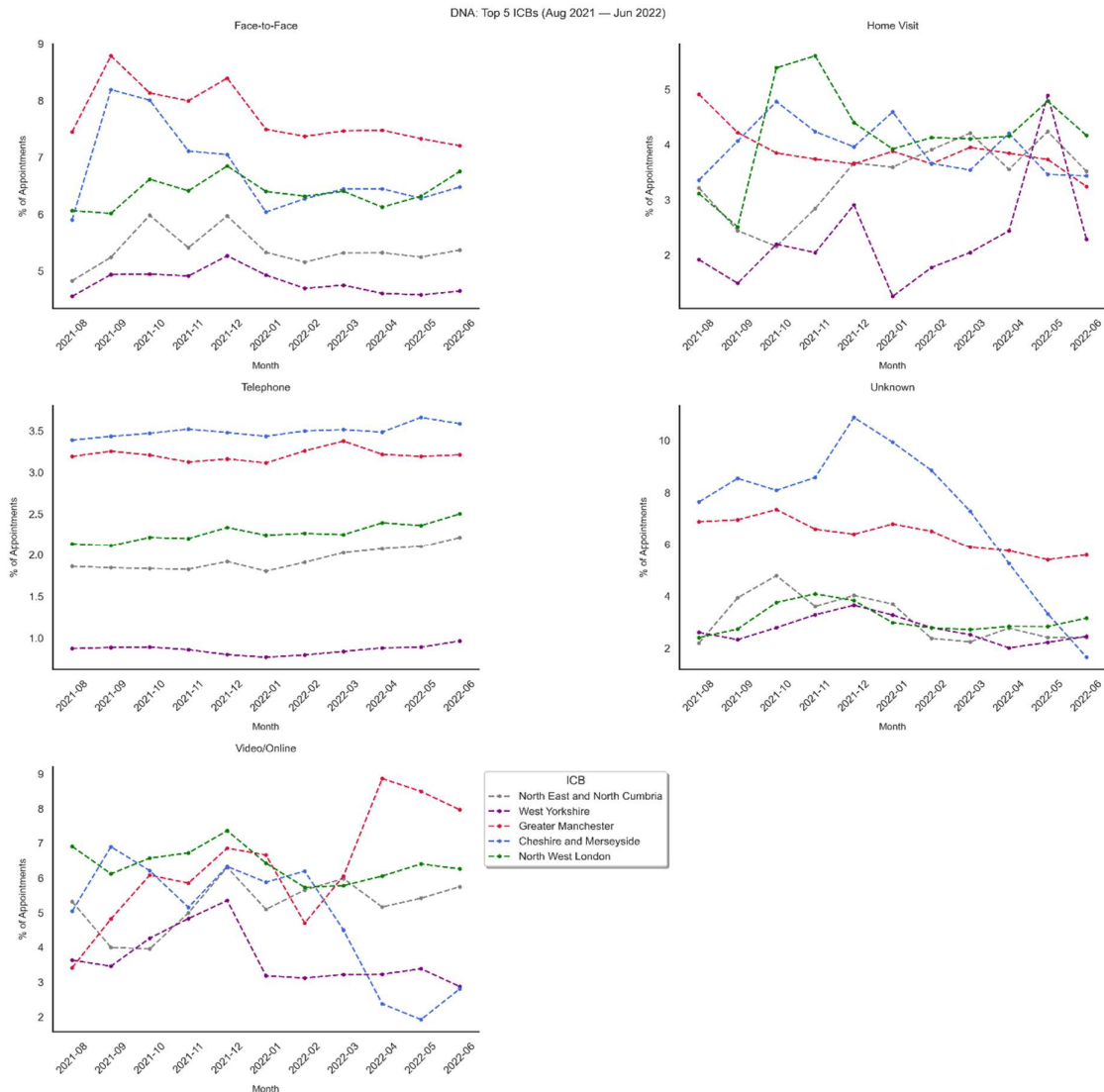
We can see that optimal capacity can be achieved by utilising weekend days. The suggestion would be to make Saturdays a normal working day (as a minimum, Sunday as well would be even better), for this I would anticipate the need for a 20% minimum increase in staffing levels to cover the additional hours, and the potential introduction of a shift rota to provide cover.

The months with the highest number of appointments also had the highest number of DNA's. It could be that DNA appointments having to be re-booked pushes up the appointment count and artificially inflates the figures.



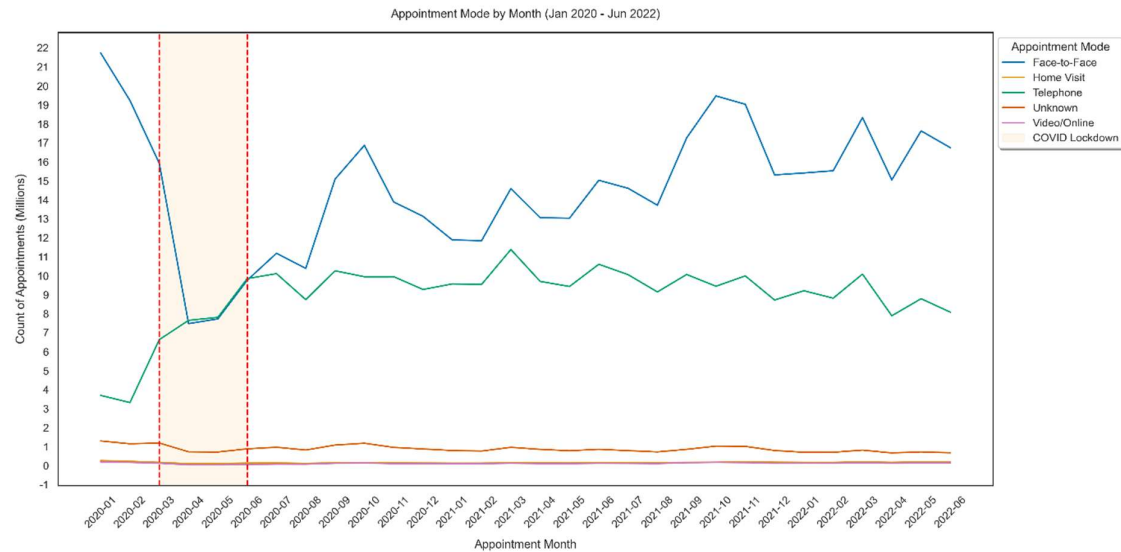
DNA's seem particularly bad in London and the North-West. Further investigation required into the cause.

Most DNA's are face-to-face appointments suggesting patients are having difficulty getting to appointments or have not been reminded that they have an appointment booked in a suitable manner. Replacing Face-to-Face with Telephone appointments where appropriate, and a better reminder system may help.

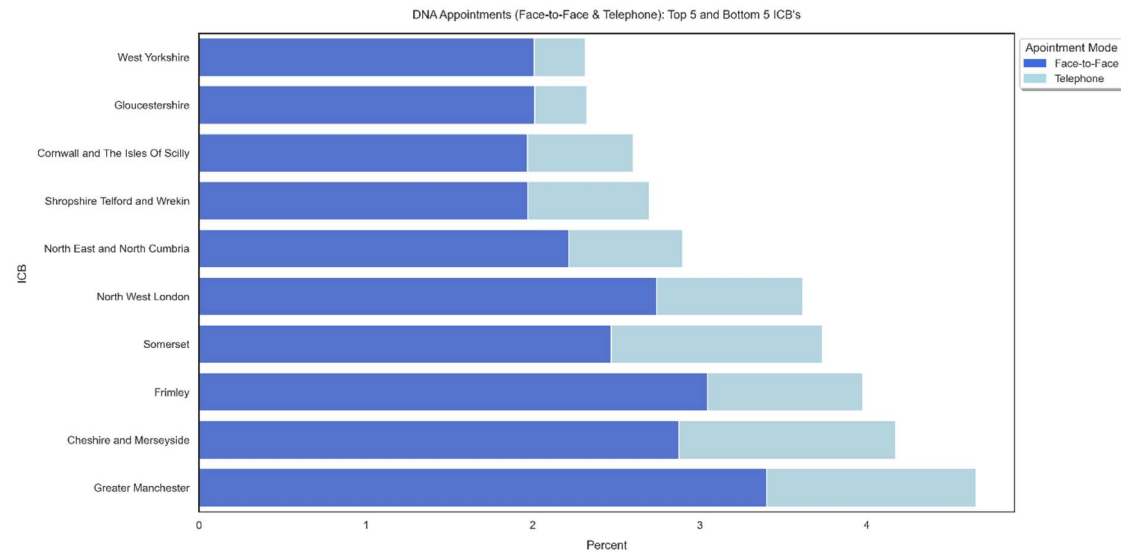


Video/Online and Home visits are small in terms of appointment numbers but have a disproportionately high DNA rate. This needs further investigation. I suggest that video/online appointments are unpopular or there are technical issues at play. I would advise these be dropped in favour of telephone appointments if possible.

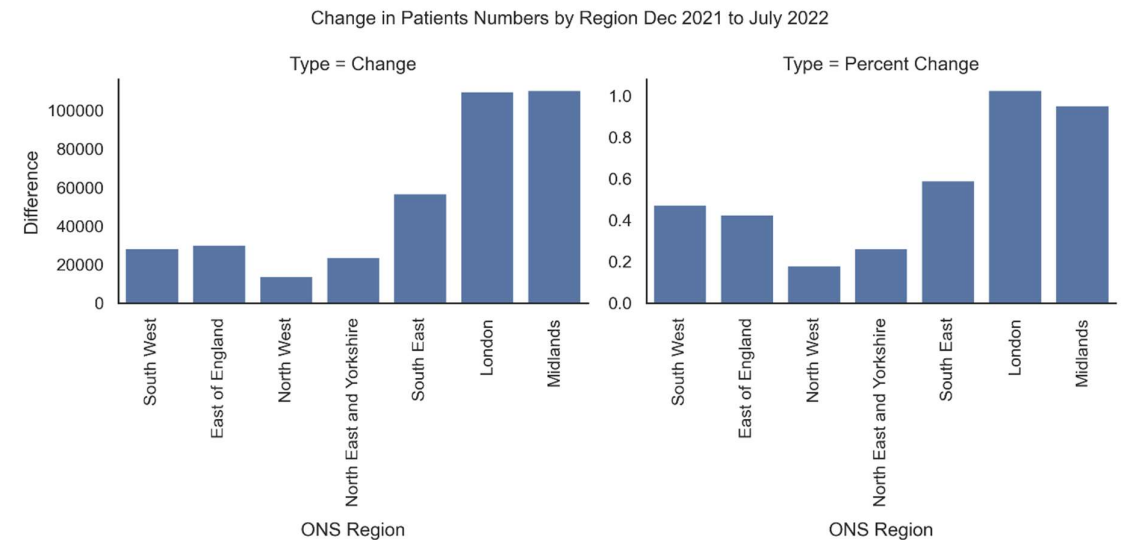
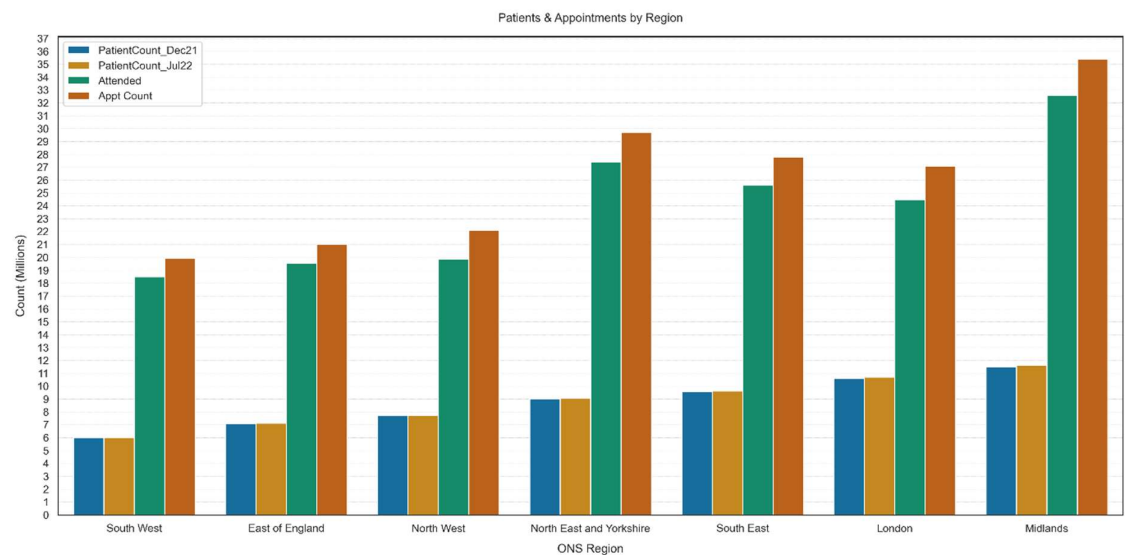
An exploratory look at appointment modes by month over the entire date range available shows a clear picture of how COVID altered the picture, with many face-to-face appointments replaced by Telephone. Telephone appointments are being dropped again in favour of face-to-face, I do not think this is a good thing, and telephone usage should be used more in appropriate situations



The following graph enforces the rationale for this. Face-to-face appointments make up a higher proportion of DNA's across a range of ICB's.



The following 2 graphs show that patient numbers are highest, and increasing most rapidly in London and the Midlands, the rate of increase being roughly 1%, around 100,000 patients per region. This is likely to exacerbate an already struggling capacity situation long term, however, more patients is not necessarily translating to more appointments as can be seen by the high appointment count in the North-East and Yorkshire relative to patient numbers.



Finally, a quick look at health care providers by region. Overall, the spread of appointments appears good. Regional variations show that GP's in London handle around 60% of appointments, whereas in the Noth East and Yorkshire this is around 45%. Other Practice staff seem to pick up more workload during peak times.

